Historic England Interview

Audio Transcript

My name's Dr. Hannah Fluck. I am the head of environmental strategy at Historic England. We provide advice and support for the heritage sector about how to look after and maintain heritage assets so archaeological sites, buildings, parks and gardens, wrecks, and how to adapt those for use, how to think about those within planning decision making. We support quite a range of different organisations with different interests and needs from owners, homeowners and occupiers, owners of assets through to those organisations whose business is heritage or conservation, and also local authorities and those who make decisions about heritage, providing guidance and frameworks for those decisions, as well as advising government on how to consider heritage in its policies.

The climate change issues that Historic England and our sector are concerned with it's pretty broad, actually. We've been thinking in quite general terms at the moment about the trends that we're seeing and likely to see in our climate. Mostly thinking about changes in precipitation. We know that we will see more intense rainfall in particular and we know that that's already causing problems, particularly for some of our older buildings. Gutters and rainwater systems are failing. People are seeing increased incidents of localised flash flooding, and a lot of the external fabric of buildings may be being penetrated by rain in a way that it hasn't done previously. We're already seeing some of those changes, and we know that's going to get worse.

We also have some concerns around ground stability. So sometimes that's connected to that change in the rainfall, sometimes to the change in temperature, but particularly issues with shrink-swell in soils, which are very rich in clay, which can cause structural problems as the ground expands and contracts more than a lot of those structures have been used to, but also ground stability with landslip and exacerbated rates of erosion, which can destroy archaeological sites and cause problems with structural integrity. Also linked to coastal erosion, which is not always just driven by the wave action in the sea, but also the ground stability of those cliff lines. And we have a lot of heritage around the coast which can be affected by that.

I did a bit of an experiment. I asked colleagues, which do you think are the primary hazards that we need to be responding to? It's quite a common approach in climate adaptation workshops to seek thoughts from the participants, from the experts in a particular field, what they think the biggest issues are. And most people, I've observed, they might think about coastal erosion or they might think about storms. Those are usually pretty high, maybe flooding. Those are the sorts of things that people would say. When we compared the National Heritage List for England against the data we had – the maps, information about climate hazards – we found that by far the highest correlation, coincidence, exposure of the assets in that list was to overheating. I think it was 73% but around three quarters of designated heritage assets in England will be exposed to the highest levels of overheating by 2060 and currently none of them are exposed to those higher levels of overheating.

But sometimes we don't know what some of those changes might mean. We know that things are getting hotter and we have more issues with overheating. We don't really know what that will mean for a lot of our archaeological sites or necessarily for some of our green heritage, our landscapes and historic places. We need to be mindful that we know these things are changing but we don't always know exactly what impact that might have.

Historic England's contribution to understanding and addressing some of these risks is something that we're, I think it's fair to say we're still developing. I often say to people that this is new for pretty much everybody. We haven't addressed climate change before. We are addressing climate change challenges now and new information is coming to light, which makes that easier in some ways. There isn't a, "this is how you do this" road map - most of us are, I mean if I'm honest, making it up as we go along to the best of our abilities and using the knowledge that we have, the expertise we have, in those areas that we're working in. That means that we are looking into what we can learn about those climate hazards, and what we know about them using the climate projections.

We're very fortunate in the UK, we have some world-class climate projections, but how you actually use those and make those accessible to people who aren't climate scientists and don't have time to become climate scientists, but still need to understand them in order to make decisions, is quite challenging. You need to know what is changing where because it isn't uniform across the whole country. There is some variation in which aspects of our climate are changing and what those impacts would be, how those will play out in different parts of the UK. We've been trying to map those hazards. We've used sustainability and data experts 3Keel to develop mapping the National Trust, first to use them to develop some hazard mapping, which proved to be pretty useful for giving an overview of how things will change, and identifying the sort of priority hazards that we need to be thinking about and planning for, and identifying gaps in our knowledge to then invest in understanding better. So that's been an important partnership.

The data that they've been using has been publicly available data from a variety of sources: from the Met Office, I think from Ordnance Survey, from BGS, from others. And there's a lot out there, the Environment Agency, for instance. But it is also a minefield. There's a huge variation, some of those data are quite old. So having somebody who can help navigate and stitch together the data that you need to answer the questions that you have is, I think, a worthwhile investment.

As we've identified the geographic distribution of some of those hazards, we're also trying to understand the vulnerability of our interests to those so we can then work out the risk. The risk is separate from the hazards. You could have, for instance, overheating. We know that just under three quarters of the National Heritage List for England assets will be exposed to very high levels of overheating by 2060 compared to none exposed to high levels of overheating currently. That's a huge change, but that might not matter: we need to try and understand which of those assets would be vulnerable to that hazard and to that exposure.

We have used climate projections for a high emissions scenario for the hazard mapping, which I referred to before, where we've been mapping at a 5km grid, the climate hazards that are most pertinent to heritage assets and their management. But what people need to know in order to take action, that's quite a coarse scale that we've used in order to map some of those hazards. It's just meant to be an indication of whether or not you should be worrying whether you need to start thinking in more detail about some of those hazards.

And then you need to think about those tipping points, what it is about that hazard that is concerning you. Does it matter the level of detail that you might need about that, in which case you probably do need to start thinking about which climate projections are most suitable given your ability to cope or not with the level of risk. How certain do you need to be and how much does it matter that an extreme thing might happen, even if it's not incredibly likely to happen?

Understanding how to respond is a whole other challenge as well because it's very hard to respond if you haven't really understood what the issue is in the first place. We're trying to do a lot of things at once when we talk about climate adaptation because we're trying to understand complex climate projection information in a way that's simple enough for people to know whether they need to worry about it, to understand how vulnerable our assets are, our interests are, to those changes, of which things we need to worry about, and then think about what we can do about them and whether we need. So in the case of Heritage, one at the extreme end of that adaptation, there will be places and sites of historic interest which we cannot adapt, which we are going to lose because of these climate change processes. For us, part of thinking about that adaptation is also preparing for what happens when we can't address those challenges and respond to those hazards in a way that's going to enable us to keep those heritage assets going. And we've been undertaking some research with the University of Exeter and the National Trust that's been supported by the AHRC, that's been looking at how we can plan for that in a positive and proactive way rather than just sort of sitting and waiting for something to happen and going, "Oh whoops, it's gone. We didn't know that was going to happen!" If we can prepare for that, then we can find ways to record those sites, to talk to people about what will happen to them and maybe even find ways to manage them that can give benefits to people and to wildlife and support that change in a positive way rather than just walking away from them.

The sorts of information that the people we work with need to plan ahead is probably, is pretty varied. It'll depend on the nature of their interest in the historic environment, which heritage assets they are either looking after or making decisions about. But for the most part, they need to know what will change. We are asking them to make decisions that are looking after heritage for future generations. We think quite long-term about things. A lot of the things that we are helping in looking after have been there for thousands of years, certainly hundreds of years, and we want them to be there for hundreds and thousands of years more.

The other thing is that people want to know what are the, the kind of no-regret options. There are certain things that we know make a huge difference to the ability of places and sites and structures and indeed people to, to help them respond and be resilient to the changes in the climate, and one of the most important is maintenance. It's a really boring topic in many ways, but it's really not, it's absolutely essential. We are trying to find ways to really explain how important it is, the good maintenance of buildings and places, ensuring that your gutters are cleared, that your brickwork has been pointed, that you've got a well-maintained place and property, and that applies to our parks and gardens and designed landscapes, that your trees are looked after and healthy. That's going to make the biggest difference to whether or not your places and these heritage assets are going to be able to be resilient to some of these changes.

We also know that we need to think holistically about the decisions we're making. At the moment, there's an awful lot of pressure to reduce greenhouse gas emissions. Quite rightly, it's a really important action which we all need to take. But if in doing so, we're reducing the ability of our places to cope with the climate changes that are coming, it's going to end up costing us more carbon in the long run. It's not going to actually be a helpful thing to do, and helping people to understand the connection between the decisions they might be making to help reduce carbon and the potential for mal-adaptation or unintended negative consequences in the future, is something else that we're trying to look at. And that can be pretty challenging because the way in which people think about making decisions is often quite compartmentalised.

Now, the other side of it is thinking about risk and about the timeframe over which we plan, and the responsibilities that we have to heritage, often described as an unrenewable resource. The archaeological sites that are buried in the ground from thousands of years ago cannot be replaced once they have gone so we have a really high level of responsibility to look after those for future generations. And we also need to think very long-term about those places. They've been there for thousands of years, we want them to be there for thousands of years more. So in that instance, the level of risk that we can accommodate is actually quite low for those sorts of places and sites. We need to really think about the worst case scenarios. My advice is usually to use high emissions scenarios, those worst case scenario projections in order to base our information on because the risk of us getting it wrong, if we used any more optimistic scenarios, we could risk making decisions which actually aren't going to secure our heritage for future generations.

And there's also another question: the point in time at which those pathways start to diverge in terms of the impact that they will have. And I think for an awful lot of people planning for the next couple of decades is all that they can really cope with thinking about. And over that timeframe, the difference is much less. It's when you get into the much longer term, over the next century, that you get that divergence. Are you going to have time sufficient in the next couple of decades to adapt the decisions that you're making now in terms of how those assets would be looked after over the coming century? Probably. In which case knowing enough to take some action to address those trends is probably sufficient.

I'm not sure if that's an answer that I'm meant to be giving when people talk about climate data, but I guess my main message is: don't get overly concerned about the details unless you know why you need to, because it's more important to be thinking about change and the direction and the trends in that change and what that means for you and your business and your interests than it is to get bogged down in the detail of which projection might be currently the most accurate.